

University of Dhaka

Department of Computer Science and Engineering, Faculty of Engineering & Technology

CSE 2206: Microcontroller and Embedded System

Assignment 3 — BME280 Environmental Sensor Interface (SPI & I2C)

LAB QUIZ — 100 MCQ QUESTION BANK

With Answer Key, CLO / Bloom's Level Mapping and Detailed Explanations

Course Code	CSE 2205 / CSE 2206	Reference Assignment	Lab Assignment 3 (BME280 SPI & I2C)
Total Questions	100	Marks per Question	1 mark
Negative Marking	None	Reference Source	BME280 Datasheet (Rev 1.24) + Problem Statement
Answer Sheet	OMR (HB / 2B pencil)	Duration (suggested)	90 minutes

OBE Alignment (Course Learning Outcomes, mapped from the Assignment 3 problem statement)

CLO	Statement	Bloom's Level	Qs Covering
CLO1	Configure SPI peripheral registers (bare-metal) on STM32 and interface BME280 as SPI slave.	2 (Apply)	32
CLO2	Configure I2C peripheral registers (bare-metal) on STM32 and interface BME280 as I2C slave.	2 (Apply)	19
CLO3	Read BME280 calibration data and apply manufacturer compensation formulas and I2C protocols.	4 (Analyze)	29
CLO4	Implement identical sensor interfaces using STM32 HAL / CubeIDE and compare both approaches.	5 (Evaluate)	8
CLO5	Use GTKterm to verify UART output and perform systematic hardware-level debugging.	3 (Apply)	12

Instructions: (1) 100 questions; 1 mark each; no negative marking. (2) Every question is drawn directly from the Assignment 3 problem statement (SPI & I2C bare-metal / HAL interfacing of the BME280) and cross-verified against the official Bosch Sensortec BME280 datasheet (Document BST-BME280-DS001-24, Revision 1.24, Feb 2024). (3) Mark answers on the OMR sheet using HB or 2B pencil only. (4) No mobile phones or electronic devices, except where explicitly permitted by the invigilator for open-datasheet sessions. (5) Section headers indicate the topic cluster; questions are numbered continuously 1–100.

PART I — QUESTIONS

SECTION 1 — BME280 Datasheet Fundamentals (Registers, Modes, Calibration)

- What value is read from BME280 register 0xD0 ("id") on a working, powered-up sensor?
(A) 0x56 (B) 0x58
(C) 0x60 (D) 0x77
- Which register and value combination performs a complete power-on-reset (soft reset) of the BME280?
(A) 0xE0, value 0xB6 (B) 0xD0, value 0x60
(C) 0xF4, value 0xB6 (D) 0xE0, value 0xFF
- In register 0xF4 (ctrl_meas), which bit positions hold the mode[1:0] field that selects the sensor's power mode?
(A) Bits 7,6,5 (B) Bits 4,3,2
(C) Bits 1,0 (D) Bit 0 only
- Per Table 25 of the BME280 datasheet, what does mode[1:0] = 11 select?
(A) Sleep mode (B) Forced mode
(C) Normal mode (D) Reserved / invalid
- What does mode[1:0] = 00 select on the BME280?
(A) Sleep mode (default after reset, lowest power) (B) Forced mode
(C) Normal mode (D) Self-test mode

6. Which register field must be written to select the humidity oversampling setting, `osrs_h[2:0]`?
- (A) 0xF2 (`ctrl_hum`) (B) 0xF4 (`ctrl_meas`)
(C) 0xF5 (`config`) (D) 0xF3 (`status`)
7. A write to `ctrl_hum` (0xF2) only takes effect after which subsequent action?
- (A) A soft reset via register 0xE0 (B) A write to `ctrl_meas` (0xF4)
(C) A write to `config` (0xF5) (D) A power cycle of VDD
8. For any BME280 oversampling field (`osrs_t`, `osrs_p`, or `osrs_h`), what does the value 000 mean?
- (A) Oversampling $\times 1$ (B) The measurement is skipped and its output register is set to 0x8000/0
(C) Oversampling $\times 16$ (maximum) (D) Invalid; the sensor rejects the write
9. For oversampling fields `osrs_t/osrs_p/osrs_h`, what does the encoded value 101 (or any other unused code) select?
- (A) Oversampling $\times 4$ (B) Oversampling $\times 16$
(C) Skipped measurement (D) Oversampling $\times 8$
10. The assignment's Indoor Navigation profile writes 0x57 to `ctrl_meas` (0xF4). What settings does this encode?
- (A) `osrs_t`= $\times 1$, `osrs_p`= $\times 1$, mode=Forced (B) `osrs_t`= $\times 2$, `osrs_p`= $\times 16$, mode=Normal
(C) `osrs_t`= $\times 4$, `osrs_p`= $\times 8$, mode=Forced (D) `osrs_t`= $\times 16$, `osrs_p`= $\times 2$, mode=Sleep
11. The assignment writes 0x10 to `config` (0xF5). What IIR filter coefficient does this select?
- (A) Filter off (B) 2
(C) 4 (D) 16
12. For the same `config` byte (0x10), what standby time `t_sb` does the encoding select?
- (A) 0.5 ms (B) 62.5 ms
(C) 10 ms (D) 1000 ms
13. Which register range holds the raw (uncompensated) pressure ADC output, `up[19:0]`?
- (A) 0xF7–0xF9 (B) 0xFA–0xFC
(C) 0xFD–0xFE (D) 0x88–0x9F
14. Which register range holds the raw temperature ADC output, `ut[19:0]`?
- (A) 0xF7–0xF9 (B) 0xFA–0xFC
(C) 0xFD–0xFE (D) 0xE1–0xE7
15. Which register range holds the raw humidity ADC output, `uh[15:0]`?
- (A) 0xFD–0xFE (B) 0xFA–0xFC
(C) 0xF7–0xF9 (D) 0xA1
16. Starting a burst read at register 0xF7 and continuing to 0xFE reads how many bytes, covering which measurements?
- (A) 6 bytes: pressure and temperature only (B) 8 bytes: pressure, temperature, and humidity
(C) 5 bytes: temperature and humidity only (D) 4 bytes: pressure only
17. Why does the BME280 datasheet strongly recommend a single burst read instead of separately addressing each data register?
- (A) Only to save power (B) To prevent a mix-up of bytes belonging to different measurements (data corruption)
(C) Because individually addressed reads are not supported (D) Because burst reads are only possible over SPI, not I2C
18. Where are the temperature calibration coefficients `dig_T1–dig_T3` stored in NVM?
- (A) 0x88–0x8D (B) 0x8E–0x9F
(C) 0xA1 (D) 0xE1–0xE7
19. Where are the pressure calibration coefficients `dig_P1–dig_P9` stored?
- (A) 0x88–0x8D (B) 0x8E–0x9F
(C) 0xA1 and 0xE1–E7 (D) 0xFA–0xFC

20. The humidity calibration coefficients dig_H1–dig_H6 are split across which register groups?
- (A) 0x88–0x9F only (B) 0xA1 (dig_H1) and 0xE1–0xE7 (dig_H2–dig_H6)
- (C) 0xFD–0xFE (D) 0xF2 and 0xF4
21. What is the BME280's specified full-accuracy operating pressure range?
- (A) 0–2000 hPa (B) 300–1100 hPa
- (C) 500–1500 hPa (D) 900–1100 hPa only
22. What is the typical response time ($\tau_{63\%}$) of the BME280 humidity element to a step change?
- (A) 1 second (B) 10 ms
- (C) 100 ms (D) 5 seconds

SECTION 2 — Compensation Formulas, t_{fine} and Data Formats

23. What is the purpose of the " t_{fine} " variable used across the BME280 compensation functions?
- (A) It stores the final filtered pressure reading (B) It carries a fine-resolution temperature value forward into the pressure compensation function
- (C) It is simply the raw ADC temperature value (D) It configures the IIR filter coefficient
24. BME280_compensate_T_int32() returns temperature with what resolution, and what does an output value of "5123" represent?
- (A) Resolution 0.1°C; means 512.3°C (B) Resolution 0.01°C; means 51.23°C
- (C) Resolution 1°C; means 5123°C (D) Resolution 0.001°C; means 5.123°C
25. In BME280_compensate_P_int64(), the returned pressure is in which fixed-point format?
- (A) Q16.16 (B) Q24.8 (24 integer bits, 8 fractional bits)
- (C) Plain integer Pa with no fraction (D) Q22.10
26. In bme280_compensate_H_int32(), the returned humidity value is in which fixed-point format?
- (A) Q22.10 (22 integer bits, 10 fractional bits) (B) Q24.8
- (C) Plain integer %RH (D) Q16.16
27. A compensated pressure output of 24674867 (Q24.8 format) corresponds to what pressure in hPa (per the datasheet's own worked example)?
- (A) 246.74 hPa (B) 963.862 hPa
- (C) 96.386 hPa (D) 24.674 hPa
28. What datatype does the datasheet recommend for best possible pressure-compensation accuracy, and why?
- (A) 8-bit signed char, to save RAM (B) 64-bit signed integer (BME280_S64_t), because 32-bit-only math loses precision
- (C) Single-precision float, mandatory on all platforms (D) 16-bit unsigned integer
29. To reconstruct the 20-bit raw pressure value adc_P from the three burst-read bytes raw[0..2] (press_msb, press_lsb, press_xlsb), which expression is correct?
- (A) (raw[0]<<12) | (raw[1]<<4) | (raw[2]>>4) (B) (raw[0]<<8) | (raw[1]<<4) | raw[2]
- (C) (raw[0]<<16) | (raw[1]<<8) | raw[2] (D) (raw[0]<<4) | (raw[1]<<12) | raw[2]
30. To reconstruct the 16-bit raw humidity value adc_H from raw[6] (hum_msb) and raw[7] (hum_lsb), which expression is correct?
- (A) (raw[6]<<8) | raw[7] (B) (raw[6]<<4) | raw[7]
- (C) (raw[6]<<12) | raw[7] (D) raw[6] | (raw[7]<<8)

SECTION 3 — SPI Protocol and Part A (Bare-Metal & HAL)

31. In the assignment's hardware wiring table, which STM32 pin is used as SPI2 MOSI to drive BME280 SDI?
- (A) PC1 (B) PC2
- (C) PC7 (D) PB9

- 32.** Which STM32 pin is used as SPI2 MISO to receive data from BME280 SDO?
- (A) PC1 (B) PC2
(C) PC7 (D) PA2
- 33.** Which STM32 pin provides the SPI2 serial clock (SCK) to the BME280?
- (A) PC1 (B) PC2
(C) PC7 (D) PB9
- 34.** Which pin is configured as a plain GPIO output and used as the active-low Chip Select (CS) for BME280 in Part A?
- (A) PA2 (B) PB9
(C) PC1 (D) PC7
- 35.** What SPI clock mode (CPOL/CPHA) does the assignment configure to talk to the BME280?
- (A) CPOL=1, CPHA=1 (Mode 11) (B) CPOL=0, CPHA=0 (Mode 00)
(C) CPOL=1, CPHA=0 (D) CPOL=0, CPHA=1
- 36.** In SPI Mode 00, on which SCK edge does the BME280 latch data present on its SDI pin?
- (A) Falling edge (B) Rising edge
(C) Both edges (double data rate) (D) Not edge-triggered; level-triggered
- 37.** How is a BME280 register WRITE command encoded on the SPI control byte?
- (A) reg_addr | 0x80 (MSB=1) (B) reg_addr & 0x7F (MSB=0)
(C) reg_addr XOR 0xFF (D) reg_addr + 1
- 38.** How is a BME280 register READ command encoded on the SPI control byte?
- (A) reg_addr & 0x7F (B) reg_addr | 0x80 (MSB=1)
(C) reg_addr - 0x80 (D) reg_addr << 1
- 39.** In SPI mode, how many bits of the full 8-bit BME280 register address are actually transmitted, and why?
- (A) 8 bits, unchanged (B) Only 7 bits, because the MSB is replaced by the read/write flag
(C) 6 bits, with 2 reserved bits (D) 16 bits, address is sent twice
- 40.** To read from BME280 register 0xF7 over SPI, what control byte is actually transmitted?
- (A) 0x77 (B) 0xF7
(C) 0x7F (D) 0x87
- 41.** In the bare-metal SPI2_CR1 configuration snippet, what does setting the MSTR bit (bit 2) accomplish?
- (A) Selects an 8-bit data frame (B) Configures the STM32 as SPI master (drives SCK and NSS logic)
(C) Enables hardware CRC (D) Sets the baud-rate prescaler
- 42.** In the bare-metal SPI2_CR1 configuration, SSM and SSI are both set to 1. What does this achieve?
- (A) Enables hardware NSS output automatically toggled by the peripheral (B) Enables software slave management, forcing the internal NSS signal
(C) Selects a 16-bit data frame format (D) Selects LSB-first bit order
- 43.** What SPI baud-rate divider (BR[2:0] = 010) is configured in the assignment's bare-metal SPI2 snippet?
- (A) fPCLK/2 (B) fPCLK/4
(C) fPCLK/8 (D) fPCLK/16
- 44.** In the SPI_TxRx() bare-metal function, which status flag must be polled before writing a new byte into SPI2->DR?
- (A) RXNE (Receive buffer not empty) (B) TXE (Transmit buffer empty)
(C) BSY (Busy flag) (D) OVR (Overrun error)
- 45.** In SPI_TxRx(), which status flag confirms a received byte is ready to be read from SPI2->DR?
- (A) TXE (B) RXNE (Receive buffer not empty)
(C) MODF (Mode fault) (D) CRCERR

46. In `BME280_SPI_ReadRegs()`, what dummy byte value is transmitted on MOSI purely to generate clock pulses while capturing data on MISO?
- (A) 0x00 (B) 0xAA
(C) 0xFF (D) 0x80
47. What is the maximum SPI clock input frequency supported by the BME280, per Table 34 of the datasheet?
- (A) 1 MHz (B) 4 MHz
(C) 10 MHz (D) 45 MHz
48. In `BME280_HAL_SPI_ReadRegs()`, what must happen to the CS (PB9) pin immediately before `HAL_SPI_Transmit()` sends the read command byte?
- (A) CS is driven HIGH to select the sensor (B) CS is driven LOW to select the sensor
(C) CS is left floating (D) CS is toggled twice
49. In the 4-wire SPI connection diagram, which BME280 pin becomes tri-stated (hi-Z) if 3-wire mode were instead selected?
- (A) SDI (B) SDO
(C) SCK (D) CSB
50. If Test A1 in Part A reads Chip ID = 0xFF instead of the expected 0x60, what is the most likely root cause per the assignment's troubleshooting note?
- (A) The IIR filter coefficient is set incorrectly (B) The sensor is not responding — check MOSI/MISO wiring
(C) The UART baud rate is mismatched (D) TIM6 is not generating interrupts

SECTION 4 — I2C Protocol and Part B (Bare-Metal & HAL)

51. In the assignment's hardware wiring table, which STM32 pin is used as I2C1 SCL?
- (A) PB6 (B) PB7
(C) PC1 (D) PC7
52. Which STM32 pin is used as I2C1 SDA?
- (A) PB6 (B) PB7
(C) PA2 (D) PA3
53. What alternate function number is assigned to I2C1 on pins PB6/PB7 in this assignment?
- (A) AF4 (B) AF5
(C) AF7 (D) AF9
54. Which GPIO output type MUST be selected for the I2C1 SCL/SDA pins, and why?
- (A) Push-pull, for maximum switching speed (B) Open-drain — mandatory for correct I2C bus operation
(C) Analog mode, to reduce coupling noise (D) Push-pull with an internal pull-down
55. What external components does the assignment specify for SDA and SCL, which many breakout boards already include on-board?
- (A) 100 Ω series resistors (B) Pull-up resistors, typically 4.7 k Ω to 3.3 V
(C) Decoupling capacitors only (D) Pull-down resistors to GND
56. What is the BME280's 7-bit I2C slave address when the SDO/ADDR pin is tied to GND?
- (A) 0x76 (B) 0x77
(C) 0xEC (D) 0xEE
57. What is the BME280's 7-bit I2C slave address when SDO is tied to VDDIO?
- (A) 0x76 (B) 0x77
(C) 0xED (D) 0xEF
58. What is the 8-bit WRITE address byte corresponding to the 7-bit address 0x76?
- (A) 0xEC (B) 0xED
(C) 0xEE (D) 0xEF

- 59.** What is the 8-bit READ address byte corresponding to the 7-bit address 0x76?
- (A) 0xEC (B) 0xED
(C) 0xEE (D) 0xEF
- 60.** Which sequence correctly describes an I2C “register-then-read” transaction with the BME280, as specified in Section B1.2?
- (A) START → addr+W → reg_addr → STOP (register address) → START → addr+W → reg_addr → RESTART → addr+R → data byte → STOP
(B) START → addr+W → reg_addr → STOP
(C) START → addr+R → reg_addr → STOP (D) START → reg_addr → addr+W → STOP
- 61.** What bus condition is defined by a HIGH-to-LOW transition on SDA while SCL is HIGH?
- (A) STOP condition (B) START condition
(C) ACK bit (D) NACK bit
- 62.** What bus condition is defined by a LOW-to-HIGH transition on SDA while SCL is HIGH?
- (A) START condition (B) STOP condition
(C) Repeated START (D) A data bit equal to 1
- 63.** In the bare-metal I2C1 configuration (Step B3.2), what value is written into I2C1->CR2 (the FREQ field) for an APB1 clock of 45 MHz?
- (A) 100 (B) 45
(C) 90 (D) 400
- 64.** For 100 kHz Standard-mode I2C with APB1 = 45 MHz, what CCR value does the assignment's own calculation produce?
- (A) 45 (B) 100
(C) 225 (D) 450
- 65.** What TRISE value does the assignment compute for Standard-mode I2C at APB1 = 45 MHz?
- (A) 45 (B) 46
(C) 100 (D) 225
- 66.** In I2C_WriteReg(), which status flag must be observed immediately after setting the START bit in I2C1->CR1?
- (A) BTF (Byte Transfer Finished) (B) SB (Start Bit generated), in SR1
(C) ADDR flag (D) STOPF flag
- 67.** After the slave address byte is sent in I2C_WriteReg()/I2C_ReadRegs(), which flag confirms the address was acknowledged by the BME280?
- (A) SB (B) ADDR, in SR1
(C) TXE (D) RXNE
- 68.** Why must I2C1->SR2 be read immediately after I2C1->SR1's ADDR flag is observed set?
- (A) To retrieve the received data byte (B) Because the specific read sequence SR1 then SR2 is what clears the ADDR flag
(C) To manually generate the STOP condition (D) To re-enable acknowledgment for the next byte
- 69.** In I2C_ReadRegs(), what special handling is applied to the very last byte of a multi-byte burst read?
- (A) The ACK bit is set to keep the bus alive (B) The ACK bit is cleared (generating NACK) before that last byte is clocked out
(C) A STOP condition is sent before the final byte is read (D) An extra dummy write is issued first
- 70.** What 8-bit device address macro does the HAL wrapper use for a BME280 with SDO tied to GND?
- (A) 0x76 (B) 0xEC
(C) (0x76 << 1) = 0xEC (D) (0x77 << 1)
- 71.** Which single HAL function call most conveniently combines “write the register pointer” and “read back N bytes” in one operation?
- (A) HAL_I2C_Master_Transmit() (B) HAL_I2C_Mem_Read()
(C) HAL_I2C_Master_Receive() (D) HAL_I2C_IsDeviceReady()

72. If Test B1 (Chip ID over I2C) fails, which of the following is the troubleshooting checklist item given in Section B6.1?

- (A) Check the TIM6 prescaler value
- (B) Check SDA/SCL pull-ups, the I2C address (0x76 vs 0x77), and that C
- (C) Check the IIR filter coefficient
- (D) Check the USART2 baud rate divisor

SECTION 5 — STM32 Clock, GPIO, USART2 and TIM6 Configuration

73. What PLL parameters does the assignment use, sourced from HSE, to reach SYSCLK = 180 MHz?

- (A) PLLM=4, PLLN=180, PLLP=2
- (B) PLLM=8, PLLN=336, PLLP=4
- (C) PLLM=2, PLLN=90, PLLP=2
- (D) PLLM=4, PLLN=360, PLLP=4

74. What Flash ACR wait-state (LATENCY) setting does the assignment require for 180 MHz operation?

- (A) 2 wait states
- (B) 3 wait states
- (C) 5 wait states
- (D) 7 wait states

75. What APB1 prescaler is applied in the clock tree so that APB1 = 45 MHz from a 180 MHz SYSCLK?

- (A) /1
- (B) /2
- (C) /4
- (D) /8

76. What APB2 prescaler is applied in the same clock tree, and what resulting frequency does it give?

- (A) /1 → 180 MHz
- (B) /2 → 90 MHz
- (C) /4 → 45 MHz
- (D) /8 → 22.5 MHz

77. Which RCC_CFGR field must be polled after enabling the PLL to confirm SYSCLK has actually switched to the PLL output?

- (A) PLLRDY
- (B) SWS[1:0] = 10
- (C) HSERDY
- (D) A dedicated CFGR lock bit

78. Which alternate function number is used for the USART2 TX/RX pins (PA2/PA3) in this assignment?

- (A) AF4
- (B) AF5
- (C) AF7
- (D) AF9

79. What baud rate is used for all UART/GTKterm communication throughout both Part A and Part B?

- (A) 9600
- (B) 19200
- (C) 57600
- (D) 115200

80. Per Step A3.3, what DIV_Mantissa and DIV_Fraction values does the assignment specify for USART2 at 115200 baud with APB1 = 45 MHz?

- (A) Mantissa=390, Fraction=10
- (B) Mantissa=24, Fraction=7
- (C) Mantissa=45, Fraction=0
- (D) Mantissa=156, Fraction=0

81. Which USART2_CR1 bits are enabled together for this lab's UART configuration?

- (A) UE only
- (B) UE, TE, RE, and RXNEIE
- (C) UE and the stop-bit field only
- (D) TE only, since only transmission is used

82. Given a required TIM6 PSC of 9000 to produce a 0.1 ms tick from a 90 MHz APB1 timer clock, what ARR value yields exactly a 1-second periodic interrupt?

- (A) ARR = 10000
- (B) ARR = 1000
- (C) ARR = 100000
- (D) ARR = 90

83. Which bit in TIM6->CR1 must be set to actually start the counter running?

- (A) UDIS
- (B) URS
- (C) CEN (Counter Enable)
- (D) ARPE

84. Which bit in TIM6->DIER must be enabled so an interrupt fires on every timer update (overflow) event?

- (A) UIE (Update Interrupt Enable)
- (B) CC1IE
- (C) TIE
- (D) UDE

85. Inside the TIM6 interrupt service routine, what must be done first, before calling BME280_ReadAll() and the UART send routine?

- (A) Disable the counter (clear CEN)
- (B) Clear the UIF (update interrupt flag) in TIM6->SR
- (C) Reload ARR with a new value
- (D) Reset the SPI/I2C peripheral

86. Which RCC register/bits must be set to enable the clocks for GPIOA, GPIOB, and GPIOC used throughout this assignment?

- (A) RCC_APB2ENR bits 0,1,2
- (B) RCC_AHB1ENR bits 0,1,2
- (C) RCC_APB1ENR bits 0,1,2
- (D) RCC_AHB2ENR bits 0,1,2

SECTION 6 — HAL vs Bare-Metal and Verification Tests

87. In Test A5 / Test B5 (HAL vs Bare-Metal comparison), what maximum temperature difference between the two implementations counts as a PASS?

- (A) ≤ 0.1 °C
- (B) ≤ 1.0 °C
- (C) ≤ 5 hPa
- (D) ≤ 5 %RH

88. What is the correct procedure specified for Test A5/B5 to fairly compare bare-metal and HAL implementations?

- (A) Run both firmwares simultaneously on two separate boards
- (B) Flash the bare-metal firmware, record three readings; then flash the HAL firmware, record three readings
- (C) Only compare the final compensated pressure, ignoring temperature
- (D) Use a spreadsheet macro to auto-generate readings without hardware

89. What is the expected UART message and pass condition for Test A1 (Chip ID check over SPI)?

- (A) "[A1] ChipID=0x60", pass condition id == 0x60
- (B) "[A1] ChipID=0xFF", pass condition id == 0xFF
- (C) "[A1] SPI OK" with no ID check
- (D) "[A1] ChipID=0x76", pass condition id == 0x76

90. What is the purpose of Test B2 ("I2C ACK Check") in Part B?

- (A) To measure the I2C bus clock frequency directly
- (B) To confirm the slave acknowledges its address after START, by polling I2C_SR1 & I2C_SR2
- (C) To verify the IIR filter coefficient
- (D) To measure UART transmission latency

91. What is checked, and what UART messages are produced, in Test A3 / B3 ("TIM6 Heartbeat")?

- (A) A free-running counter incremented once per TIM6 ISR call
- (B) The SPI clock frequency, not by a timer, but by a scope
- (C) The I2C CCR register value printed once at boot
- (D) The number of BME280 soft resets performed

92. In Test A4/B4 (Sensor Plausibility), what condition must ALL be true simultaneously for a PASS to be reported?

- (A) temp_C in [15,40], pres_hPa in [900,1100], hum_RH in [0,100]
- (B) temp_C in [0,100], pres_hPa in [0,2000], hum_RH in [0,50]
- (C) Only temp_C needs to be checked
- (D) Only pres_hPa needs to be checked

93. Per the troubleshooting note in Section A7, what two hardware items should be checked if all SPI sensor readings show 0.00?

- (A) USART2 baud rate and TIM6 prescaler
- (B) The CS pin (PB9) and the SPI mode (CPOL=0, CPHA=0)
- (C) The I2C pull-up resistors
- (D) The PLL PLLN multiplier

94. Per the troubleshooting note in Section B7, what is described as "the most common I2C fault" if readings are stuck at 0 or show garbage values?

- (A) A wrong TIM6 ARR value
- (B) Missing pull-up resistors on SDA/SCL
- (C) An incorrect USART2 stop-bit setting
- (D) Using 4-wire instead of 3-wire SPI

SECTION 7 — GTKterm, Unit Conversions and Final SPI/I2C Comparison

95. Which command grants a Linux user permanent permission to access /dev/ttyACM0 without running GTKterm as root?

- (A) chmod 777 /dev/ttyACM0
- (B) sudo usermod -aG dialout \$USER (then log out/in)
- (C) sudo chown root /dev/ttyACM0
- (D) sudo apt install ttyACM0

96. What GTKterm port settings must match the firmware's USART2 configuration for correct communication?

- (A) 9600 baud, 7 data bits, even parity, 2 stop bits
- (B) 115200 baud, 8 data bits, no parity, 1 stop bit, no flow control
- (C) 38400 baud, 8 data bits, odd parity, 1 stop bit
- (D) 115200 baud, 9 data bits, no parity, 1.5 stop bits

97. What formula does the assignment specify for converting compensated temperature from °C to °F?

(A) $T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times 9/5 + 32$

(B) $T(^{\circ}\text{F}) = T(^{\circ}\text{C}) + 32$

(C) $T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times 5/9 + 32$

(D) $T(^{\circ}\text{F}) = (T(^{\circ}\text{C}) - 32) \times 9/5$

98. What formula does the assignment specify for converting pressure from Pa to hPa?

(A) $P(\text{hPa}) = P(\text{Pa}) \times 100$

(B) $P(\text{hPa}) = P(\text{Pa}) / 100$

(C) $P(\text{hPa}) = P(\text{Pa}) / 1000$

(D) $P(\text{hPa}) = P(\text{Pa}) \times 0.1$

99. According to the assignment's final SPI-vs-I2C comparison table, which statement is TRUE?

(A) SPI requires pull-up resistors while I2C does not

(B) I2C uses only 2 wires and does not need a per-slave Chip Select, while SPI does

(C) I2C supports full-duplex communication while SPI does not

(D) SPI's maximum practical bus speed is lower than I2C's Fast mode

100. Why does the BME280 produce numerically identical sensor readings regardless of whether it is accessed via SPI (Part A) or I2C (Part B)?

(A) Because SPI and I2C use different ADCs inside the sensor

(B) Because the underlying sensing elements, ADC, and internal registers are the same

(C) Because the STM32 firmware silently converts I2C data into SPI format

(D) SPI and I2C are guaranteed by the datasheet to produce identical readings before compensation

PART II — ANSWER KEY (Instructor Copy)

Quick-reference answer key with CLO and Bloom's Level mapping. Full worked explanations follow in Part III.

Q	Ans	CLO	Bloom	Q	Ans	CLO	Bloom
1	C	CLO3	L3	51	A	CLO2	L3
2	A	CLO1	L3	52	B	CLO2	L3
3	C	CLO3	L3	53	A	CLO2	L3
4	C	CLO3	L3	54	B	CLO2	L3
5	A	CLO3	L3	55	B	CLO2	L3
6	A	CLO3	L3	56	A	CLO2	L3
7	B	CLO1	L3	57	B	CLO2	L3
8	B	CLO3	L3	58	A	CLO2	L3
9	B	CLO3	L3	59	B	CLO2	L3
10	B	CLO3	L4	60	B	CLO2	L3
11	D	CLO3	L4	61	B	CLO2	L3
12	A	CLO3	L4	62	B	CLO2	L3
13	A	CLO3	L3	63	B	CLO2	L3
14	B	CLO3	L3	64	C	CLO2	L4
15	A	CLO3	L3	65	B	CLO2	L4
16	B	CLO3	L3	66	B	CLO2	L3
17	B	CLO5	L4	67	B	CLO2	L3
18	A	CLO3	L3	68	B	CLO2	L4
19	B	CLO3	L3	69	B	CLO2	L3
20	B	CLO3	L3	70	C	CLO4	L3
21	B	CLO3	L3	71	B	CLO4	L3
22	A	CLO3	L3	72	B	CLO5	L3
23	B	CLO3	L4	73	A	CLO1	L3
24	B	CLO3	L4	74	C	CLO1	L3
25	B	CLO3	L4	75	C	CLO1	L3
26	A	CLO3	L4	76	B	CLO1	L3
27	B	CLO3	L4	77	B	CLO1	L3
28	B	CLO4	L5	78	C	CLO1	L3
29	A	CLO3	L3	79	D	CLO5	L3
30	A	CLO3	L3	80	B	CLO1	L3
31	A	CLO1	L3	81	B	CLO1	L3
32	B	CLO1	L3	82	A	CLO1	L4
33	C	CLO1	L3	83	C	CLO1	L3
34	B	CLO1	L3	84	A	CLO1	L3
35	B	CLO1	L3	85	B	CLO1	L3
36	B	CLO1	L3	86	B	CLO1	L3
37	B	CLO1	L3	87	A	CLO4	L5
38	B	CLO1	L3	88	B	CLO4	L5
39	B	CLO1	L3	89	A	CLO5	L3
40	B	CLO1	L3	90	B	CLO5	L3

Q	Ans	CLO	Bloom	Q	Ans	CLO	Bloom
41	B	CLO1	L3	91	A	CLO5	L3
42	B	CLO1	L3	92	A	CLO5	L4
43	C	CLO1	L3	93	B	CLO5	L3
44	B	CLO1	L3	94	B	CLO5	L3
45	B	CLO1	L3	95	B	CLO5	L3
46	C	CLO1	L3	96	B	CLO5	L3
47	C	CLO3	L3	97	A	CLO3	L3
48	B	CLO4	L3	98	B	CLO3	L3
49	B	CLO1	L4	99	B	CLO4	L5
50	B	CLO5	L3	100	B	CLO4	L5

Answer string (grouped in blocks of 5, in question order 1–100):

CACCA ABBBB DAABA BBABB BABBB ABBAA ABCBB BBBBB BBCBB CBBBB
 ABABB ABABB BBBCB BBBBC BBACC BBCDB BACAB BABAB ABBBB BABBB

PART III — DETAILED EXPLANATIONS (Worked Solutions for All 100 Questions)

Each explanation cites the exact clause of the Assignment 3 problem statement and/or the BME280 datasheet (document BST-BME280-DS001-24) that justifies the correct answer.

SECTION 1 — BME280 Datasheet Fundamentals (Registers, Modes, Calibration)

Q1. What value is read from BME280 register 0xD0 ("id") on a working, powered-up sensor?

Correct Answer: (C) 0x60 | CLO: CLO3 | Bloom: Apply (L3) | Topic: Chip ID

Explanation: Datasheet Section 5.4.1: register 0xD0 "id" contains chip_id[7:0], which is fixed at 0x60 for the BME280 (this differs from the BMP280, whose id reads 0x56/0x57/0x58).

Q2. Which register and value combination performs a complete power-on-reset (soft reset) of the BME280?

Correct Answer: (A) 0xE0, value 0xB6 | CLO: CLO1 | Bloom: Apply (L3) | Topic: Soft Reset

Explanation: Datasheet Section 5.4.2: the "reset" register is at 0xE0. Writing 0xB6 triggers the complete power-on-reset procedure; any other value has no effect and the readout is always 0x00.

Q3. In register 0xF4 (ctrl_meas), which bit positions hold the mode[1:0] field that selects the sensor's power mode?

Correct Answer: (C) Bits 1,0 | CLO: CLO3 | Bloom: Apply (L3) | Topic: ctrl_meas Register

Explanation: Table 22 of the datasheet: ctrl_meas (0xF4) bit7-5 = osrs_t[2:0], bit4-2 = osrs_p[2:0], and bit1-0 = mode[1:0].

Q4. Per Table 25 of the BME280 datasheet, what does mode[1:0] = 11 select?

Correct Answer: (C) Normal mode | CLO: CLO3 | Bloom: Apply (L3) | Topic: Sensor Modes

Explanation: mode[1:0]: 00 = Sleep, 01 and 10 = Forced, 11 = Normal (perpetual cycling of measurement and standby periods) — this is the mode used by the assignment's Indoor Navigation configuration.

Q5. What does mode[1:0] = 00 select on the BME280?

Correct Answer: (A) Sleep mode (default after reset, lowest power) | CLO: CLO3 | Bloom: Apply (L3) | Topic: Sensor Modes

Explanation: Sleep mode is entered by default after power-on-reset; no measurements occur, but all registers (including calibration data) remain accessible.

Q6. Which register field must be written to select the humidity oversampling setting, osrs_h[2:0]?

Correct Answer: (A) 0xF2 (ctrl_hum) | CLO: CLO3 | Bloom: Apply (L3) | Topic: ctrl_hum Register

Explanation: Datasheet Section 5.4.3: register 0xF2 "ctrl_hum" bits 2:0 hold osrs_h[2:0], controlling humidity oversampling.

Q7. A write to ctrl_hum (0xF2) only takes effect after which subsequent action?

Correct Answer: (B) A write to ctrl_meas (0xF4) | CLO: CLO1 | Bloom: Apply (L3) | Topic: ctrl_hum Register

Explanation: The datasheet explicitly notes: "Changes to this register only become effective after a write operation to ctrl_meas." This is why the assignment's init sequence writes 0xF2 before 0xF4.

Q8. For any BME280 oversampling field (osrs_t, osrs_p, or osrs_h), what does the value 000 mean?

Correct Answer: (B) The measurement is skipped and its output register is set to 0x8000/0x80000 | CLO: CLO3 | Bloom:

Apply (L3) | Topic: Oversampling

Explanation: Tables 20, 23 and 24 all show 000 = "Skipped (output set to 0x8000 / 0x80000)" for humidity/pressure/temperature respectively.

Q9. For oversampling fields osrs_t/osrs_p/osrs_h, what does the encoded value 101 (or any other unused code) select?

Correct Answer: (B) Oversampling ×16 | CLO: CLO3 | Bloom: Apply (L3) | Topic: Oversampling

Explanation: Tables 20/23/24: 001=×1, 010=×2, 011=×4, 100=×8, and 101 ("others") = ×16, the maximum oversampling level.

Q10. The assignment's Indoor Navigation profile writes 0x57 to ctrl_meas (0xF4). What settings does this encode?

Correct Answer: (B) osrs_t=×2, osrs_p=×16, mode=Normal | CLO: CLO3 | Bloom: Analyze (L4) | Topic: Register Encoding

Explanation: 0x57 = 0b01010111. Splitting: osrs_t[7:5]=010(×2), osrs_p[4:2]=101(×16), mode[1:0]=11(Normal) — exactly the CSE2206 Indoor-Navigation profile in Section 4.3 of the problem statement.

Q11. The assignment writes 0x10 to config (0xF5). What IIR filter coefficient does this select?

Correct Answer: (D) 16 | CLO: CLO3 | Bloom: Analyze (L4) | Topic: Register Encoding

Explanation: 0x10 = 0b00010000. filter[4:2] = 100 = coefficient 16, t_sb[7:5] = 000 = 0.5 ms standby, spi3w_en (bit0) = 0 (4-wire SPI).

Q12. For the same config byte (0x10), what standby time t_{sb} does the encoding select?

Correct Answer: (A) 0.5 ms | CLO: CLO3 | Bloom: Analyze (L4) | Topic: Register Encoding

Explanation: $t_{sb}[7:5] = 000$ maps to 0.5 ms per Table 27 of the datasheet — the shortest standby time, appropriate for the high output-data-rate Indoor Navigation profile.

Q13. Which register range holds the raw (uncompensated) pressure ADC output, $up[19:0]$?

Correct Answer: (A) 0xF7–0xF9 | CLO: CLO3 | Bloom: Apply (L3) | Topic: Data Registers

Explanation: Registers $press_msb$, $press_lsb$, $press_xlsb$ occupy 0xF7–0xF9 and hold the 20-bit raw pressure value.

Q14. Which register range holds the raw temperature ADC output, $ut[19:0]$?

Correct Answer: (B) 0xFA–0xFC | CLO: CLO3 | Bloom: Apply (L3) | Topic: Data Registers

Explanation: $temp_msb$, $temp_lsb$, $temp_xlsb$ occupy 0xFA–0xFC and hold the 20-bit raw temperature value.

Q15. Which register range holds the raw humidity ADC output, $uh[15:0]$?

Correct Answer: (A) 0xFD–0xFE | CLO: CLO3 | Bloom: Apply (L3) | Topic: Data Registers

Explanation: hum_msb and hum_lsb occupy 0xFD–0xFE, holding the 16-bit raw humidity value.

Q16. Starting a burst read at register 0xF7 and continuing to 0xFE reads how many bytes, covering which measurements?

Correct Answer: (B) 8 bytes: pressure, temperature, and humidity | CLO: CLO3 | Bloom: Apply (L3) | Topic: Burst Read

Explanation: 0xF7–0xFE spans $press_msb \dots hum_lsb = 8$ consecutive bytes, giving all three uncompensated measurements in one transaction.

Q17. Why does the BME280 datasheet strongly recommend a single burst read instead of separately addressing each data register?

Correct Answer: (B) To prevent a mix-up of bytes belonging to different measurements (data shadowing) and reduce interface traffic | CLO: CLO5 | Bloom: Analyze (L4) | Topic: Data Shadowing

Explanation: Section 4.1 explains that in Normal mode a new measurement can complete mid-readout; shadow registers guarantee consistency only if all data registers are read in one burst, ending on the SPI CSB rising edge or the I2C STOP condition.

Q18. Where are the temperature calibration coefficients dig_T1 – dig_T3 stored in NVM?

Correct Answer: (A) 0x88–0x8D | CLO: CLO3 | Bloom: Apply (L3) | Topic: Calibration Data

Explanation: Table 16: dig_T1 (unsigned short) at 0x88/0x89, dig_T2 and dig_T3 (signed shorts) at 0x8A–0x8D.

Q19. Where are the pressure calibration coefficients dig_P1 – dig_P9 stored?

Correct Answer: (B) 0x8E–0x9F | CLO: CLO3 | Bloom: Apply (L3) | Topic: Calibration Data

Explanation: dig_P1 (unsigned short) through dig_P9 (signed shorts) occupy 0x8E–0x9F, sixteen consecutive bytes for nine 16-bit words.

Q20. The humidity calibration coefficients dig_H1 – dig_H6 are split across which register groups?

Correct Answer: (B) 0xA1 (dig_H1) and 0xE1–0xE7 (dig_H2 – dig_H6) | CLO: CLO3 | Bloom: Apply (L3) | Topic: Calibration Data

Explanation: dig_H1 (unsigned char) is at 0xA1; dig_H2 – dig_H6 occupy 0xE1–0xE7, with $dig_H4/H5$ sharing nibbles of 0xE4/0xE5/0xE6 in a packed 12-bit format.

Q21. What is the BME280's specified full-accuracy operating pressure range?

Correct Answer: (B) 300–1100 hPa | CLO: CLO3 | Bloom: Apply (L3) | Topic: Sensor Specs

Explanation: Table 3 (Pressure sensor specification): full-accuracy operating pressure range is 300 to 1100 hPa (300...1100 hPa) between 0...+65 °C.

Q22. What is the typical response time ($\tau_{63\%}$) of the BME280 humidity element to a step change?

Correct Answer: (A) 1 second | CLO: CLO3 | Bloom: Apply (L3) | Topic: Sensor Specs

Explanation: Table 2 (Humidity parameter specification): response time to complete 63% of a step (0→90 or 90→0 %RH, 25°C) is typically 1 s.

SECTION 2 — Compensation Formulas, t_{fine} and Data Formats

Q23. What is the purpose of the "t_fine" variable used across the BME280 compensation functions?

Correct Answer: (B) It carries a fine-resolution temperature value forward into the pressure and humidity compensation formulas | CLO: CLO3 | Bloom: Analyze (L4) | Topic: t_fine

Explanation: The datasheet states: "The variable t_fine (signed 32 bit) carries a fine resolution temperature value over to the pressure and humidity compensation formula." It must be computed by BME280_compensate_T_int32() before compensating P or H.

Q24. BME280_compensate_T_int32() returns temperature with what resolution, and what does an output value of "5123" represent?

Correct Answer: (B) Resolution 0.01°C; means 51.23°C | CLO: CLO3 | Bloom: Analyze (L4) | Topic: Compensation Output

Explanation: The datasheet comment reads: "Returns temperature in DegC, resolution is 0.01 DegC. Output value of '5123' equals 51.23 DegC." The assignment then divides by 100.0 to get °C for UART printout.

Q25. In BME280_compensate_P_int64(), the returned pressure is in which fixed-point format?

Correct Answer: (B) Q24.8 (24 integer bits, 8 fractional bits) | CLO: CLO3 | Bloom: Analyze (L4) | Topic: Compensation Output

Explanation: Comment: "Returns pressure in Pa as unsigned 32 bit integer in Q24.8 format... Output value of '24674867' represents 24674867/256 = 96386.2 Pa." The assignment divides by 256 then by 100 to obtain hPa.

Q26. In bme280_compensate_H_int32(), the returned humidity value is in which fixed-point format?

Correct Answer: (A) Q22.10 (22 integer bits, 10 fractional bits) | CLO: CLO3 | Bloom: Analyze (L4) | Topic: Compensation Output

Explanation: Comment: "Returns humidity in %RH as unsigned 32 bit integer in Q22.10 format... Output value of '47445' represents 47445/1024 = 46.333 %RH." The assignment divides by 1024.0 to obtain %RH.

Q27. A compensated pressure output of 24674867 (Q24.8 format) corresponds to what pressure in hPa (per the datasheet's own worked example)?

Correct Answer: (B) 963.862 hPa | CLO: CLO3 | Bloom: Analyze (L4) | Topic: Worked Example

Explanation: 24674867 / 256 = 96386.2 Pa = 963.862 hPa — taken verbatim from the datasheet's compensation function comment.

Q28. What datatype does the datasheet recommend for best possible pressure-compensation accuracy, and why?

Correct Answer: (B) 64-bit signed integer (BME280_S64_t), because 32-bit-only math loses roughly 1 Pa accuracy at high filter settings | CLO: CLO4 | Bloom: Evaluate (L5) | Topic: Compensation Accuracy

Explanation: Section 8.2 notes the 32-bit fallback "has an accuracy of typically 1 Pa (1-sigma)" and adds significant noise at high filter levels; 64-bit integer math (used by BME280_compensate_P_int64) is preferred whenever the MCU supports it, as the Cortex-M4 in this lab does.

Q29. To reconstruct the 20-bit raw pressure value adc_P from the three burst-read bytes raw[0..2] (press_msb, press_lsb, press_xlsb), which expression is correct?

Correct Answer: (A) (raw[0]<<12) | (raw[1]<<4) | (raw[2]>>4) | CLO: CLO3 | Bloom: Apply (L3) | Topic: ADC Reconstruction

Explanation: Section A3.7/B3.4 of the problem statement gives exactly this expression: the MSB is shifted left by 12, the LSB by 4, and only the top 4 bits of the XLSB byte are used (>>4), reconstructing the full 20-bit value.

Q30. To reconstruct the 16-bit raw humidity value adc_H from raw[6] (hum_msb) and raw[7] (hum_lsb), which expression is correct?

Correct Answer: (A) (raw[6]<<8) | raw[7] | CLO: CLO3 | Bloom: Apply (L3) | Topic: ADC Reconstruction

Explanation: Humidity has only two data bytes (16-bit, no XLSB), so adc_H = (raw[6]<<8) | raw[7], per Section A3.7 of the problem statement.

SECTION 3 — SPI Protocol and Part A (Bare-Metal & HAL)

Q31. In the assignment's hardware wiring table, which STM32 pin is used as SPI2 MOSI to drive BME280 SDI?

Correct Answer: (A) PC1 | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Pinout

Explanation: Table A2 (Hardware Connections — SPI): MOSI (Data Out) = PC1 → BME280 SDI.

Q32. Which STM32 pin is used as SPI2 MISO to receive data from BME280 SDO?

Correct Answer: (B) PC2 | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Pinout

Explanation: Table A2: MISO (Data In) = PC2 → BME280 SDO.

Q33. Which STM32 pin provides the SPI2 serial clock (SCK) to the BME280?

Correct Answer: (C) PC7 | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Pinout

Explanation: Table A2: SCK (Clock) = PC7 → BME280 SCK.

Q34. Which pin is configured as a plain GPIO output and used as the active-low Chip Select (CS) for BME280 in Part A?

Correct Answer: (B) PB9 | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Pinout

Explanation: Table A2: CS (Chip Select) = PB9, configured as a GPIO Output (not an SPI alternate function), driven LOW to select and HIGH to deselect the BME280.

Q35. What SPI clock mode (CPOL/CPHA) does the assignment configure to talk to the BME280?

Correct Answer: (B) CPOL=0, CPHA=0 (Mode 00) | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Mode

Explanation: Section A1.1: CPOL=0 (SCK idle LOW), CPHA=0 (data sampled on the first/rising edge) — SPI Mode 00, described as the BME280's default and lowest-power mode.

Q36. In SPI Mode 00, on which SCK edge does the BME280 latch data present on its SDI pin?

Correct Answer: (B) Rising edge | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Timing

Explanation: Datasheet Section 6.3: "Data on SDI is latched by the device at SCK rising edge and SDO is changed at SCK falling edge."

Q37. How is a BME280 register WRITE command encoded on the SPI control byte?

Correct Answer: (B) reg_addr & 0x7F (MSB=0) | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Read/Write Encoding

Explanation: Section A1.2: "Write: send (reg_addr & 0x7F) — MSB = 0." The most significant bit of the SPI control byte acts as the R/W flag; 0 = write.

Q38. How is a BME280 register READ command encoded on the SPI control byte?

Correct Answer: (B) reg_addr | 0x80 (MSB=1) | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Read/Write Encoding

Explanation: Section A1.2: "Read: send (reg_addr | 0x80) — MSB = 1; address auto-increments for burst reads."

Q39. In SPI mode, how many bits of the full 8-bit BME280 register address are actually transmitted, and why?

Correct Answer: (B) Only 7 bits, because the MSB is replaced by the read/write flag | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Addressing

Explanation: Section 6.3: "Only 7 bits of the register addresses are used; the MSB of register address is not used and replaced by a read/write bit."

Q40. To read from BME280 register 0xF7 over SPI, what control byte is actually transmitted?

Correct Answer: (B) 0xF7 | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Addressing Example

Explanation: Datasheet example: "address 0xF7 is accessed using SPI register address 0x77. For write access, the byte 0x77 is transferred, for read access, the byte 0xF7 is transferred" (0x77 | 0x80 = 0xF7).

Q41. In the bare-metal SPI2_CR1 configuration snippet, what does setting the MSTR bit (bit 2) accomplish?

Correct Answer: (B) Configures the STM32 as SPI master (drives SCK and NSS logic) | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI2 Bare-Metal Config

Explanation: MSTR=1 in SPI_CR1 places the peripheral in master mode, meaning the STM32 generates SCK and controls the transaction, as required since the BME280 is always an SPI slave.

Q42. In the bare-metal SPI2_CR1 configuration, SSM and SSI are both set to 1. What does this achieve?

Correct Answer: (B) Enables software slave management, forcing the internal NSS signal HIGH so the peripheral stays in master mode | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI2 Bare-Metal Config

Explanation: SSM=1 (Software Slave Management) disables automatic hardware NSS handling; SSI=1 then manually forces the internal NSS input high so MSTR mode is not accidentally cleared. Chip select is instead handled manually via GPIO PB9.

Q43. What SPI baud-rate divider (BR[2:0] = 010) is configured in the assignment's bare-metal SPI2 snippet?

Correct Answer: (C) fPCLK/8 | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI2 Bare-Metal Config

Explanation: BR[2:0] = 2 (binary 010) selects a divide-by-8 prescaler on the SPI peripheral clock, per the standard STM32 SPI_CR1 BR field encoding used in the given register snippet.

Q44. In the SPI_TxRx() bare-metal function, which status flag must be polled before writing a new byte into SPI2->DR?

Correct Answer: (B) TXE (Transmit buffer empty) | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Bare-Metal Driver

Explanation: The code waits for SPI2->SR bit 1 (TXE) before writing SPI2->DR, ensuring the transmit shift register is ready to accept new data.

Q45. In SPI_TxRx(), which status flag confirms a received byte is ready to be read from SPI2->DR?

Correct Answer: (B) RXNE (Receive buffer not empty) | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Bare-Metal Driver

Explanation: The code waits for SPI2->SR bit 0 (RXNE) before reading SPI2->DR, since full-duplex SPI always produces one received byte for every byte transmitted.

Q46. In `BME280_SPI_ReadRegs()`, what dummy byte value is transmitted on MOSI purely to generate clock pulses while capturing data on MISO?

Correct Answer: (C) 0xFF | CLO: CLO1 | Bloom: Apply (L3) | Topic: SPI Burst Read

Explanation: The loop calls `SPI_TxRx(0xFF)` for each byte after the read command — the value itself is irrelevant to the slave since MOSI is ignored during a read; only the resulting clock edges matter.

Q47. What is the maximum SPI clock input frequency supported by the BME280, per Table 34 of the datasheet?

Correct Answer: (C) 10 MHz | CLO: CLO3 | Bloom: Apply (L3) | Topic: SPI Electrical Spec

Explanation: Table 34 (SPI timings): `F_spi` (SPI clock input frequency) ranges 0 to 10 MHz.

Q48. In `BME280_HAL_SPI_ReadRegs()`, what must happen to the CS (PB9) pin immediately before `HAL_SPI_Transmit()` sends the read command byte?

Correct Answer: (B) CS is driven LOW to select the sensor | CLO: CLO4 | Bloom: Apply (L3) | Topic: HAL SPI Driver

Explanation: `HAL_GPIO_WritePin(GPIOB, GPIO_PIN_9, GPIO_PIN_RESET)` pulls CS LOW first (active-low chip select), exactly mirroring the manual CS handling used in the bare-metal driver.

Q49. In the 4-wire SPI connection diagram, which BME280 pin becomes tri-stated (hi-Z) if 3-wire mode were instead selected?

Correct Answer: (B) SDO | CLO: CLO1 | Bloom: Analyze (L4) | Topic: SPI Wiring Modes

Explanation: Table 35: in 3-wire mode SDO is “DNC” (do not connect) / hi-Z, because SDI becomes bidirectional (SDI/SDO) and carries both directions of data — the assignment specifically uses 4-wire mode with a dedicated MISO (SDO) line.

Q50. If Test A1 in Part A reads Chip ID = 0xFF instead of the expected 0x60, what is the most likely root cause per the assignment's troubleshooting note?

Correct Answer: (B) The sensor is not responding — check MOSI/MISO wiring | CLO: CLO5 | Bloom: Apply (L3) | Topic: SPI Debugging

Explanation: Section A7 states: “If chip ID = 0xFF: sensor not responding — verify MOSI/MISO wiring.” 0xFF typically indicates the MISO line is floating high with no slave driving it.

SECTION 4 — I2C Protocol and Part B (Bare-Metal & HAL)

Q51. In the assignment's hardware wiring table, which STM32 pin is used as I2C1 SCL?

Correct Answer: (A) PB6 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Pinout

Explanation: Table B2 (Hardware Connections — I2C): SCL (Clock) = PB6 → BME280 SCK.

Q52. Which STM32 pin is used as I2C1 SDA?

Correct Answer: (B) PB7 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Pinout

Explanation: Table B2: SDA (Data) = PB7 → BME280 SDI/SDO.

Q53. What alternate function number is assigned to I2C1 on pins PB6/PB7 in this assignment?

Correct Answer: (A) AF4 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Pinout

Explanation: Table B2 lists AF4 for both SCL (PB6) and SDA (PB7); Step B3.1 also instructs setting AF4 in `GPIOB_AFRL` for both pins.

Q54. Which GPIO output type MUST be selected for the I2C1 SCL/SDA pins, and why?

Correct Answer: (B) Open-drain — mandatory for correct I2C bus operation | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C GPIO Config

Explanation: Step B3.1 explicitly states: “Set `OTYPER = 1` (open-drain) for PB6 and PB7 — mandatory for I2C.” I2C is a wired-AND bus; push-pull outputs would cause bus contention.

Q55. What external components does the assignment specify for SDA and SCL, which many breakout boards already include on-board?

Correct Answer: (B) Pull-up resistors, typically 4.7 kΩ to 3.3 V | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Hardware

Explanation: Section B2 notes both SCL and SDA require external pull-up resistors (4.7 kΩ to 3.3 V) for the open-drain bus to reach a valid HIGH level; many BME280 breakout boards include these on-board.

Q56. What is the BME280's 7-bit I2C slave address when the SDO/ADDR pin is tied to GND?

Correct Answer: (A) 0x76 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Addressing

Explanation: Table B1: SDO connected to GND → 7-bit address = 0x76 (this is also the default address assumed throughout the assignment's code, e.g. `#define BME280_I2C_ADDR 0x76`).

Q57. What is the BME280's 7-bit I2C slave address when SDO is tied to VDDIO?

Correct Answer: (B) 0x77 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Addressing

Explanation: Table B1: SDO connected to VDDIO → 7-bit address = 0x77 (this is also the BMP280's fixed address, since BME280 is register-compatible with it).

Q58. What is the 8-bit WRITE address byte corresponding to the 7-bit address 0x76?

Correct Answer: (A) 0xEC | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Addressing

Explanation: 8-bit write byte = (0x76 << 1) | 0 = 0xEC, matching Table B1's "8-bit Write" column for SDO=GND.

Q59. What is the 8-bit READ address byte corresponding to the 7-bit address 0x76?

Correct Answer: (B) 0xED | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Addressing

Explanation: 8-bit read byte = (0x76 << 1) | 1 = 0xED, matching Table B1's "8-bit Read" column for SDO=GND.

Q60. Which sequence correctly describes an I2C "register-then-read" transaction with the BME280, as specified in Section B1.2?

Correct Answer: (B) START → addr+W → reg_addr → RESTART → addr+R → data byte(s) → NACK → STOP | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Transaction

Explanation: Section B1.2: the master first writes the target register address (addr+W, reg_addr), then issues a RESTART, re-addresses the slave in read mode (addr+R), clocks out the data byte(s), and terminates the final byte with NACK before STOP.

Q61. What bus condition is defined by a HIGH-to-LOW transition on SDA while SCL is HIGH?

Correct Answer: (B) START condition | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Fundamentals

Explanation: Quiz-style MCU fact reused in context: a START condition is signalled by SDA falling while SCL is held HIGH, which the BME280 I2C1->CR1 START bit generates in hardware.

Q62. What bus condition is defined by a LOW-to-HIGH transition on SDA while SCL is HIGH?

Correct Answer: (B) STOP condition | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Fundamentals

Explanation: A STOP condition is SDA rising while SCL is held HIGH; in the bare-metal driver this is generated by setting I2C1->CR1 bit 9 (STOP) after the last data byte.

Q63. In the bare-metal I2C1 configuration (Step B3.2), what value is written into I2C1->CR2 (the FREQ field) for an APB1 clock of 45 MHz?

Correct Answer: (B) 45 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C1 Bare-Metal Config

Explanation: CR2 FREQ must equal the APB1 peripheral clock expressed in MHz; the snippet sets I2C1->CR2 = 45 for the 45 MHz APB1 bus used throughout this lab.

Q64. For 100 kHz Standard-mode I2C with APB1 = 45 MHz, what CCR value does the assignment's own calculation produce?

Correct Answer: (C) 225 | CLO: CLO2 | Bloom: Analyze (L4) | Topic: I2C1 Bare-Metal Config

Explanation: Step B3.2: $T_{high} = T_{low} = 5 \mu s$ in Standard mode; $CCR = T_{high} / T_{pclk1} = 5000 \text{ ns} / (1000/45 \text{ ns}) = 225$, matching the given snippet I2C1->CCR = 225.

Q65. What TRISE value does the assignment compute for Standard-mode I2C at APB1 = 45 MHz?

Correct Answer: (B) 46 | CLO: CLO2 | Bloom: Analyze (L4) | Topic: I2C1 Bare-Metal Config

Explanation: Step B3.2: maximum rise time in Standard mode is 1000 ns, so $TRISE = (1000 \text{ ns} / T_{pclk1}) + 1 = 45 + 1 = 46$, matching I2C1->TRISE = 46 in the snippet.

Q66. In I2C_WriteReg(), which status flag must be observed immediately after setting the START bit in I2C1->CR1?

Correct Answer: (B) SB (Start Bit generated), in SR1 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Bare-Metal Driver

Explanation: After CR1 |= (1<<8) generates START, the code polls SR1 bit 0 (SB) to confirm the start condition has actually been transmitted on the bus before sending the address byte.

Q67. After the slave address byte is sent in I2C_WriteReg()/I2C_ReadRegs(), which flag confirms the address was acknowledged by the BME280?

Correct Answer: (B) ADDR, in SR1 | CLO: CLO2 | Bloom: Apply (L3) | Topic: I2C Bare-Metal Driver

Explanation: The code polls SR1 bit 1 (ADDR) after writing the address+R/W byte to DR; ADDR being set confirms the slave ACKed its address.

Q68. Why must I2C1->SR2 be read immediately after I2C1->SR1's ADDR flag is observed set?

Correct Answer: (B) Because the specific read sequence SR1 then SR2 is what clears the ADDR flag in hardware | CLO:

CLO2 | Bloom: Analyze (L4) | Topic: I2C Bare-Metal Driver

Explanation: On the STM32 I2C peripheral, ADDR is cleared only by the specific sequence: read SR1 (already done to test the flag) followed by a read of SR2. Skipping this read leaves ADDR set and stalls the peripheral.

Q69. In I2C_ReadRegs(), what special handling is applied to the very last byte of a multi-byte burst read?

Correct Answer: (B) The ACK bit is cleared (generating NACK) before that last byte is clocked in, then STOP is issued | CLO:

CLO2 | Bloom: Apply (L3) | Topic: I2C Burst Read

Explanation: The code clears CR1 bit 10 (ACK) just before reading the final byte via I2C1->DR, so the master sends NACK to signal the slave that this is the last requested byte, then generates STOP.

Q70. What 8-bit device address macro does the HAL wrapper use for a BME280 with SDO tied to GND?

Correct Answer: (C) (0x76 << 1) = 0xEC | CLO: CLO4 | Bloom: Apply (L3) | Topic: HAL I2C Driver

Explanation: The HAL wrapper defines #define BME280_ADDR (0x76 << 1), because HAL_I2C_Master_Transmit/Receive expect the address pre-shifted into 8-bit form, i.e. 0xEC.

Q71. Which single HAL function call most conveniently combines “write the register pointer” and “read back N bytes” in one operation?

Correct Answer: (B) HAL_I2C_Mem_Read() | CLO: CLO4 | Bloom: Apply (L3) | Topic: HAL I2C Driver

Explanation: BME280_HAL_I2C_MemRead() in Section B5.2 uses HAL_I2C_Mem_Read(&hi2c1, BME280_ADDR, reg, I2C_MEMADD_SIZE_8BIT, buf, len, HAL_MAX_DELAY) — a single call that internally performs the write-then-repeated-start-read sequence.

Q72. If Test B1 (Chip ID over I2C) fails, which of the following is the troubleshooting checklist item given in Section B6.1?

Correct Answer: (B) Check SDA/SCL pull-ups, the I2C address (0x76 vs 0x77), and that OTyPER is set to open-drain | CLO:

CLO5 | Bloom: Apply (L3) | Topic: I2C Debugging

Explanation: Section B6.1's “On Fail” column for Test B1 lists exactly these three checks: pull-up resistors, the 0x76-vs-0x77 address selection, and open-drain GPIO configuration.

SECTION 5 — STM32 Clock, GPIO, USART2 and TIM6 Configuration

Q73. What PLL parameters does the assignment use, sourced from HSE, to reach SYSCLK = 180 MHz?

Correct Answer: (A) PLLM=4, PLLN=180, PLLP=2 | CLO: CLO1 | Bloom: Apply (L3) | Topic: Clock Configuration

Explanation: Step A3.1 specifies PLLM=4, PLLN=180, PLLP=2 with an HSE source. With an 8 MHz HSE crystal this gives $(8/4) \times 180/2 = 180$ MHz.

Q74. What Flash ACR wait-state (LATENCY) setting does the assignment require for 180 MHz operation?

Correct Answer: (C) 5 wait states | CLO: CLO1 | Bloom: Apply (L3) | Topic: Clock Configuration

Explanation: Step A3.1: “Flash latency: FLASH_ACR LATENCY[3:0] = 0101 (5 wait states)”, required at VDD=3.3V for SYSCLK up to 180 MHz on the STM32F446.

Q75. What APB1 prescaler is applied in the clock tree so that APB1 = 45 MHz from a 180 MHz SYSCLK?

Correct Answer: (C) /4 | CLO: CLO1 | Bloom: Apply (L3) | Topic: Clock Configuration

Explanation: Step A3.1 states “AHB=/1, APB1=/4, APB2=/2”, giving APB1 = $180/4 = 45$ MHz, which is used throughout for USART2 and I2C1 timing calculations.

Q76. What APB2 prescaler is applied in the same clock tree, and what resulting frequency does it give?

Correct Answer: (B) /2 → 90 MHz | CLO: CLO1 | Bloom: Apply (L3) | Topic: Clock Configuration

Explanation: APB2 = /2 gives 90 MHz. Because the APB1 prescaler exceeds 1, the APB1 timer clock (feeding TIM6) is doubled to $2 \times 45 = 90$ MHz, which is exactly the value used in the TIM6 PSC calculation.

Q77. Which RCC_CFGR field must be polled after enabling the PLL to confirm SYSCLK has actually switched to the PLL output?

Correct Answer: (B) SWS[1:0] = 10 | CLO: CLO1 | Bloom: Apply (L3) | Topic: Clock Configuration

Explanation: Step A3.1: “Switch clock: SW[1:0]=10 in RCC_CFGR. Wait for SWS[1:0]=10” — SWS reflects the currently active system clock source and must read back 10 (PLL) to confirm the switch succeeded.

Q78. Which alternate function number is used for the USART2 TX/RX pins (PA2/PA3) in this assignment?

Correct Answer: (C) AF7 | CLO: CLO1 | Bloom: Apply (L3) | Topic: USART2 Config

Explanation: Step A3.2: “PA2, PA3 — AF7 (USART2 TX/RX)”, the standard alternate function mapping for USART2 on these pins.

Q79. What baud rate is used for all UART/GTKterm communication throughout both Part A and Part B?

Correct Answer: (D) 115200 | CLO: CLO5 | Bloom: Apply (L3) | Topic: USART2 Config

Explanation: Both the firmware (Step A3.3) and the GTKterm port configuration table (Section 3.4) specify 115200 baud, 8 data bits, no parity, 1 stop bit, no flow control.

Q80. Per Step A3.3, what DIV_Mantissa and DIV_Fraction values does the assignment specify for USART2 at 115200 baud with APB1 = 45 MHz?

Correct Answer: (B) Mantissa=24, Fraction=7 | CLO: CLO1 | Bloom: Apply (L3) | Topic: USART2 BRR

Explanation: Step A3.3 explicitly states “BRR: DIV_Mantissa=24, DIV_Fraction=7 (APB1 = 45 MHz)”, consistent with $USARTDIV = fPCLK/(16 \times \text{baud}) = 45,000,000/(16 \times 115200) \approx 24.41$.

Q81. Which USART2_CR1 bits are enabled together for this lab's UART configuration?

Correct Answer: (B) UE, TE, RE, and RXNEIE | CLO: CLO1 | Bloom: Apply (L3) | Topic: USART2 Config

Explanation: Step A3.3: “USART CR1: enable USART (bit 13), TX (bit 3), RX (bit 2), RXNEIE (bit 5)” — RX and its interrupt are enabled even though the lab mainly transmits, for completeness/future use.

Q82. Given a required TIM6 PSC of 9000 to produce a 0.1 ms tick from a 90 MHz APB1 timer clock, what ARR value yields exactly a 1-second periodic interrupt?

Correct Answer: (A) ARR = 10000 | CLO: CLO1 | Bloom: Analyze (L4) | Topic: TIM6 Configuration

Explanation: Step A3.4: PSC=9000 gives a 0.1 ms tick (90 MHz/9000=10 kHz); ARR=10000 ticks $\times$ 0.1 ms/tick = 1000 ms = 1 second between update events.

Q83. Which bit in TIM6->CR1 must be set to actually start the counter running?

Correct Answer: (C) CEN (Counter Enable) | CLO: CLO1 | Bloom: Apply (L3) | Topic: TIM6 Configuration

Explanation: CEN (Counter Enable, bit 0 of TIMx_CR1) must be set to 1 to start the timer counting; Step A3.4 sets this bit last, after PSC/ARR/DIER are configured.

Q84. Which bit in TIM6->DIER must be enabled so an interrupt fires on every timer update (overflow) event?

Correct Answer: (A) UIE (Update Interrupt Enable) | CLO: CLO1 | Bloom: Apply (L3) | Topic: TIM6 Configuration

Explanation: Step A3.4: “TIM6 DIER bit 0 (UIE) = 1” enables the update interrupt so the NVIC dispatches to the TIM6 ISR once per second.

Q85. Inside the TIM6 interrupt service routine, what must be done first, before calling BME280_ReadAll() and the UART send routine?

Correct Answer: (B) Clear the UIF (update interrupt flag) in TIM6->SR | CLO: CLO1 | Bloom: Apply (L3) | Topic: TIM6 ISR

Explanation: Step A3.4: “ISR: clear UIF, call BME280_ReadAll(), call UART_SendSensorData()”. Failing to clear UIF causes the ISR to immediately re-trigger.

Q86. Which RCC register/bits must be set to enable the clocks for GPIOA, GPIOB, and GPIOC used throughout this assignment?

Correct Answer: (B) RCC_AHB1ENR bits 0,1,2 | CLO: CLO1 | Bloom: Apply (L3) | Topic: GPIO Clock Enable

Explanation: Step A3.2: “Enable GPIO clocks: RCC_AHB1ENR bits 0 (GPIOA), 1 (GPIOB), 2 (GPIOC).” On the STM32F4 family, GPIO ports sit on the AHB1 bus, not APB.

SECTION 6 — HAL vs Bare-Metal and Verification Tests

Q87. In Test A5 / Test B5 (HAL vs Bare-Metal comparison), what maximum temperature difference between the two implementations counts as a PASS?

Correct Answer: (A) ≤ 0.1 °C | CLO: CLO4 | Bloom: Evaluate (L5) | Topic: HAL vs Bare-Metal Test

Explanation: Both comparison tables (Section A6.5 and B6.5) use the column header “Pass (≤ 0.1 °C)?” as the acceptance criterion between three bare-metal and three HAL readings.

Q88. What is the correct procedure specified for Test A5/B5 to fairly compare bare-metal and HAL implementations?

Correct Answer: (B) Flash the bare-metal firmware, record three readings; then flash the HAL firmware, record three readings; then compare | CLO: CLO4 | Bloom: Evaluate (L5) | Topic: HAL vs Bare-Metal Test

Explanation: Section A6.5: “Flash bare-metal, record three readings. Flash HAL version, record three. Fill in the table.” The two firmwares are compared sequentially on the same physical hardware.

Q89. What is the expected UART message and pass condition for Test A1 (Chip ID check over SPI)?

Correct Answer: (A) “[A1] ChipID=0x60”, pass condition id == 0x60 | CLO: CLO5 | Bloom: Apply (L3) | Topic: SPI Verification Test

Explanation: Section A6.1's table: Expected “[A1] ChipID=0x60”, Pass Condition “id == 0x60”, matching the BME280's fixed chip_id value.

Q90. What is the purpose of Test B2 (“I2C ACK Check”) in Part B?

Correct Answer: (B) To confirm the slave acknowledges its address after START, by polling the ADDR flag with a timeout and reporting ACK OK/FAIL | CLO: CLO5 | Bloom: Apply (L3) | Topic: I2C Verification Test

Explanation: Section B6.2 sends START then the slave address, and polls SR1 ADDR with a software timeout; if ADDR never sets, it reports “[B2] I2C ACK FAIL”, otherwise “[B2] I2C ACK OK” after clearing ADDR via the SR2 read.

Q91. What is checked, and what UART messages are produced, in Test A3 / B3 (“TIM6 Heartbeat”)?

Correct Answer: (A) A free-running counter incremented once per TIM6 ISR call, expected to increment exactly once per second | CLO: CLO5 | Bloom: Apply (L3) | Topic: TIM6 Verification Test

Explanation: Sections A6.3/B6.3: a static uint32_t ticks counter is incremented inside the TIM6 ISR and printed as “[A3] Tick:%lu” (or [B3]); correct operation shows the counter incrementing exactly once per second in GTKterm.

Q92. In Test A4/B4 (Sensor Plausibility), what condition must ALL be true simultaneously for a PASS to be reported?

Correct Answer: (A) temp_C in [15,40], pres_hPa in [900,1100], hum_RH in [0,100] | CLO: CLO5 | Bloom: Analyze (L4) | Topic: Plausibility Test

Explanation: The plausibility check code in Sections A6.4/B6.4 fails if temp_C<15 or >40, OR pres_hPa<900 or >1100, OR hum_RH<0 or >100; only if none of these trip does it print “Plausibility PASS.”

Q93. Per the troubleshooting note in Section A7, what two hardware items should be checked if all SPI sensor readings show 0.00?

Correct Answer: (B) The CS pin (PB9) and the SPI mode (CPOL=0, CPHA=0) | CLO: CLO5 | Bloom: Apply (L3) | Topic: SPI Debugging

Explanation: Section A7: “If all values read 0.00: check CS pin (PB9) and SPI Mode (CPOL=0, CPHA=0).” Incorrect CS control or clock polarity/phase prevents the BME280 from ever responding correctly.

Q94. Per the troubleshooting note in Section B7, what is described as “the most common I2C fault” if readings are stuck at 0 or show garbage values?

Correct Answer: (B) Missing pull-up resistors on SDA/SCL | CLO: CLO5 | Bloom: Apply (L3) | Topic: I2C Debugging

Explanation: Section B7 states verbatim: “If readings are stuck at 0 or show garbage values: check pull-up resistors on SDA/SCL. Missing pull-ups are the most common I2C fault.”

SECTION 7 — GTKterm, Unit Conversions and Final SPI/I2C Comparison

Q95. Which command grants a Linux user permanent permission to access /dev/ttyACM0 without running GTKterm as root?

Correct Answer: (B) sudo usermod -aG dialout \$USER (then log out/in) | CLO: CLO5 | Bloom: Apply (L3) | Topic: GTKterm Setup

Explanation: Section 3.3: adding the current user to the “dialout” group with usermod -aG dialout \$USER, followed by a logout/login, grants persistent serial-port access without sudo/chmod hacks.

Q96. What GTKterm port settings must match the firmware's USART2 configuration for correct communication?

Correct Answer: (B) 115200 baud, 8 data bits, no parity, 1 stop bit, no flow control | CLO: CLO5 | Bloom: Apply (L3) | Topic: GTKterm Setup

Explanation: Section 3.4's configuration table specifies Baud=115200, Bits=8, Parity=None, Stop Bits=1, Flow Control=None — matching the bare-metal USART2 CR1/CR2 setup exactly.

Q97. What formula does the assignment specify for converting compensated temperature from °C to °F?

Correct Answer: (A) $T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times 9/5 + 32$ | CLO: CLO3 | Bloom: Apply (L3) | Topic: Unit Conversion

Explanation: Section 4.4: $T(^{\circ}\text{F}) = T(^{\circ}\text{C}) \times (9.0/5.0) + 32.0$, the standard Celsius-to-Fahrenheit conversion used identically in both Part A and Part B firmware.

Q98. What formula does the assignment specify for converting pressure from Pa to hPa?

Correct Answer: (B) $P(\text{hPa}) = P(\text{Pa}) / 100$ | CLO: CLO3 | Bloom: Apply (L3) | Topic: Unit Conversion

Explanation: Section 4.4: $P(\text{hPa}) = P(\text{Pa}) / 100.0$. Combined with the Q24.8 pressure format, the full conversion from the compensation function's raw output to hPa is (raw/256)/100.

Q99. According to the assignment's final SPI-vs-I2C comparison table, which statement is TRUE?

Correct Answer: (B) I2C uses only 2 wires and does not need a per-slave Chip Select, whereas SPI uses 4 wires including CS | CLO: CLO4 | Bloom: Evaluate (L5) | Topic: SPI vs I2C Comparison

Explanation: Section 5's comparison table: SPI = 4 wires (MOSI, MISO, SCK, CS) with a required CS pin and no pull-ups; I2C = 2 wires (SDA, SCL), address-based slave selection (no CS), but requires external pull-up resistors. SPI is full-duplex and much faster (up to $f_{PCLK}/2$), while I2C is half-duplex.

Q100. Why does the BME280 produce numerically identical sensor readings regardless of whether it is accessed via SPI (Part A) or I2C (Part B)?

Correct Answer: (B) Because the underlying sensing elements, ADC, and internal register map are identical; only the bus protocol used to access those same registers differs | CLO: CLO4 | Bloom: Evaluate (L5) | Topic: Protocol Comparison

Explanation: The BME280 has one internal measurement/ADC/compensation-parameter subsystem; SPI and I2C are merely two alternative digital interfaces for reading and writing the exact same register map (Section 6, "Digital Interfaces"), so the compensation algorithm and its C code are protocol-independent.